

GITHUB LINK: <https://github.com/Kathirmanicse12/QR-authenticated-blockchain-genomic-data-sharing.git>

Step-by-Step Process to Run the Project:

Step 1: Install Required Software

Install Python, OpenCV, and necessary libraries such as NumPy and TensorFlow using pip.

Step 2: Set Up Hardware

Connect the two cameras (road-facing and driver-facing) to the system and ensure they are properly detected.

Step 3: Download Project Files

Place all project files, including trained models and scripts, into a single project folder.

Step 4: Configure Environment

Set the correct file paths for models, datasets, and camera indices in the code.

Step 5: Run Object Detection Module

Execute the script for road object detection to identify vehicles, pedestrians, and obstacles.

Step 6: Run Driver Monitoring Module

Start the driver monitoring system to detect distractions like phone usage or head movement.

Step 7: Integrate Both Modules

Run the main program that combines both camera inputs into a unified system.

Step 8: Monitor Output

Check the real-time output window displaying detections and alerts.

Step 9: Test the System

Perform testing under different scenarios such as traffic conditions and driver behaviors.

Step 10: Troubleshooting

Fix any issues related to camera input, detection accuracy, or system performance.